MATH · THE FRAME IT NEVER NEEDED

Eight variants, one frame

Every slide after this one is a math slide on the shared Form frame. The running header, the footer and page number, and the **Q3 REVIEW** mark in the masthead bay are what a chrome-exempt section could never render — `meta:` did nothing at all on a math slide until this change.

LINEAR REGRESSION · OLS

Q3 REVIEW

The hero equation and its legend, side by side.

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector
- X — design matrix, $n \times p$
- y — response vector, length n
- $X^\top X$ — Gram matrix, $p \times p$, must be invertible

CHAIN RULE · FOUR STEPS

Q3 REVIEW

The proof chain no longer pins its own heading.

$f(x+h) - f(x) = f'(x)\,h + O(h^2)$	*subtract* $f(x)$
$\frac{f(x+h)-f(x)}{h} = f'(x) + O(h)$	*divide by* $h \neq 0$
$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$	**take the limit**

CONTINUITY · IVT

Q3 REVIEW

Cards center in the stage, not under an absolute title.

DEFINITION. A function $f : [a,b] \to \mathbb{R}$ is *continuous* on $[a,b]$ if $\lim_{x \to c} f(x) = f(c)$ for every $c \in [a,b]$.

THEOREM. Let f be continuous on $[a,b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a,b)$ with $f(c) = y$.

PROOF. Set $S = \{x \in [a,b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$.

ESTIMATORS · ASYMPTOTICS

Q3 REVIEW

The columns are the stage now, so nothing spans them.

ORDINARY LEAST SQUARES

$$\hat{\beta}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top y$$

Unbiased under exogeneity; minimum variance among linear unbiased estimators.

RIDGE

$$\hat{\beta}_{\lambda} = (X^\top X + \lambda I)^{-1} X^\top y$$

Biased, lower variance. The penalty λ buys stability when $X^\top X$ is ill-conditioned.

DESIGN MATRIX · STRUCTURE

Q3 REVIEW

A wrapped property no longer returns to the left margin.

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- SHAPE — $n \times (p+1)$
- ROWS — observations
- COLS — intercept + p features
- RANK — full rank is what gives OLS a unique solution
- COLUMN 0 — all-ones, absorbs the intercept

LU · A FACTORIZATION

Q3 REVIEW

A label that is a symbol is set as a symbol.

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

- A — the original matrix being factorized
- L — lower-triangular, unit diagonal
- U — upper-triangular
- USE — solve $Ax = b$ by forward then back substitution

EFFECT SIZE · THE READING

Q3 REVIEW

The title lands where every other title in the deck lands.

$$\hat{\beta} = \mathbf{0.42} \pm \mathbf{0.03}$$

95% CI: $[0.36,\ 0.48]$

$p < 0.001 \quad \cdot \quad n = 1{,}204$

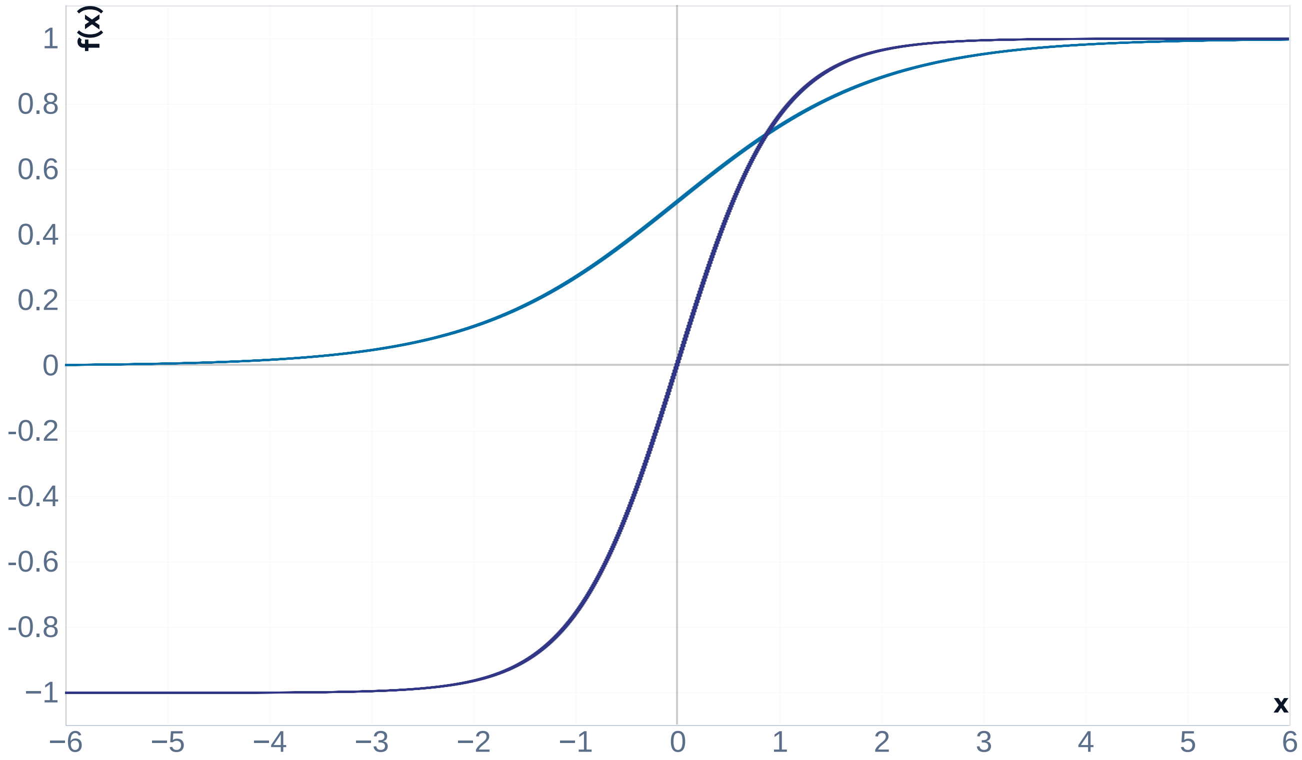
For every additional unit of exposure the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. The effect size is the headline; the p-value only rules out chance.

LOGISTIC LINK · THE SHAPE

Q3 REVIEW

The plot is sized against its cell, not against the slide.

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



Maps $\mathbb{R} \to (0,1)$. S-shaped, $\sigma(0) = 0.5$, steepest slope at the origin.